AI 101 FOR MECHANICAL ENGINEERS

MODULE 15 / LESSON 3

Responsible use and human oversight

Undergraduate course



Lesson outcomes

Identify plausible unsupported output and disclosure needs.

Specify named human review and action responsibility.

Unsupported generated output

Hallucination or confabulation is plausible false or unsupported output. It can include fabricated references, material properties and equations.

Check original sources and known numerical cases. Data coverage bias and social fairness concerns are related but different from a neural bias term. Use the intended meaning explicitly.

Privacy, disclosure and security

Do not share restricted personal, assessment or proprietary information without appropriate permission.

Follow institution rules, disclose AI help and retain your reasoning trail.

Generated code and connected tools also need security review before use in real systems.

Resources and named human responsibility

AI computation consumes physical resources.

Prefer methods that meet the task with reasonable data and compute demands. This course claims no numerical footprint for a particular tool.

Human oversight must specify who checks outputs, can stop actions and owns decisions, not merely say “human in the loop.”

Lesson vocabulary

Hallucination: Plausible but false or unsupported generated output. Example: An invented standard citation.

Human oversight: Defined human responsibility and ability to review, intervene or stop. Example: An engineer approves an operating action.

Worked exercise

What evidence and disclosure would you attach to an AI-assisted lab submission?

Pause and write an answer before moving to the explanation.

Exercise explanation

Provide data provenance, units, a clean runnable notebook, split IDs, baseline comparisons, independent checks, limitations and a disclosure of permitted AI assistance.

Concept reference table 1

Responsible-use issue	Evidence or action
Unsupported generated claim	Original-source check
Restricted information	Permission before disclosure
AI assistance	Disclosure under course policy

Teaching example. Read the table alongside the worked explanation.

Concept reference table 2

Responsible-use issue	Evidence or action
Human oversight	Named reviewer and stop authority

Teaching example. Read the table alongside the worked explanation.

Knowledge check 1

Responsible AI-assisted submissions should include…

A. Source and unit checks

B. Disclosure of permitted AI help

C. A reasoning and reproducibility trail

D. Unverified citations copied as evidence

Select every supported option.

Knowledge check 2

Human oversight should specify…

A. Who checks and can intervene

B. Only a slogan

C. That all actions are approved automatically

D. That uncertainty is hidden

Knowledge-check explanations

Check 1: A, B, C. Evidence and disclosure support accountability.

Check 2: A. Responsibility and intervention must be concrete.

Lesson recap

Identify plausible unsupported output and disclosure needs.

Specify named human review and action responsibility.

Apply the ideas to this question: What evidence and disclosure would you attach to an AI-assisted lab submission?

Sources and further reading

Primary source: AI 101 for Mechanical Engineers, Module 15, Lesson 3: Responsible use and human oversight.

Worked cases and numerical examples are teaching material. Synthetic evidence does not validate real equipment.

Google Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

scikit-learn User Guide
https://scikit-learn.org/stable/user_guide.html